

Quiz (Carolina Reaper)

- Q1.** Factor the expression $x^2 + 5x + 4x + 20$ by grouping.
(A) $x(x + 5) + 4(x + 5)$
(B) $x(x + 4) + 5(x + 4)$
(C) $x(x + 4) + 5(x + 5)$
(D) $x(x + 5) + 4(x + 4)$
(E) $x(x + 4) + 5(x + 3)$
- Q2.** A shipping company has a cost function $C(x) = x^2 + 8x + 16x + 64$ where x is the number of packages delivered. Factor $C(x)$ by grouping.
(A) $x(x + 8) + 16(x + 4)$
(B) $x(x + 16) + 8(x + 8)$
(C) $x(x + 8) + 16(x + 8)$
(D) $x(x + 16) + 8(x + 4)$
(E) $x(x + 8) + 16(x + 16)$
- Q3.** A delivery truck travels $3x + 12$ miles in the morning and $2x + 8$ miles in the afternoon. Factor the total distance traveled $D(x) = 3x + 12 + 2x + 8$ by grouping.
(A) $x(3 + 2) + (12 + 8)$
(B) $x(3 + 2) + (12 + 4)$
(C) $x(5) + (20)$
(D) $x(5) + (10)$
(E) $x(3) + (2x + 20)$
- Q4.** The cost of transporting x crates is $2x^2 + 10x + 8x + 40$. Factor the cost expression by grouping.
(A) $2x(x + 5) + 8(x + 5)$
(B) $2x(x + 4) + 10(x + 4)$
(C) $2x(x + 8) + 5(x + 8)$
(D) $2x(x + 5) + 8(x + 4)$
(E) $2x(x + 4) + 10(x + 5)$
- Q5.** Verify that $(x + 2)(x + 3)$ is factored by showing that $(x + 2)(x + 3) = x^2 + 5x + 6$.
(A) $x^2 + 5x + 5$
(B) $x^2 + 5x + 6$
(C) $x^2 + 4x + 6$
(D) $x^2 + 6x + 5$
(E) $x^2 + 6x + 6$
- Q6.** A traffic flow model is given by $F(x) = 2x^2 + 12x + 10x + 40$. Factor $F(x)$ by grouping.
(A) $2x(x + 6) + 10(x + 4)$
(B) $2x(x + 10) + 6(x + 6)$
(C) $2x(x + 6) + 10(x + 6)$
(D) $2x(x + 12) + 5(x + 8)$
(E) $2x(x + 10) + 6(x + 4)$
- Q7.** Factor the expression $x^2 + 6x + 4x + 24$ by grouping.
(A) $x(x + 6) + 4(x + 6)$
(B) $x(x + 4) + 6(x + 6)$
(C) $x(x + 10) + 2(x + 12)$
(D) $x(x + 6) + 4(x + 4)$
(E) $x(x + 4) + 6(x + 4)$
- Q8.** A package delivery company has a fuel cost function $C(x) = x^2 + 4x + 6x + 12$. Factor $C(x)$ by grouping.
(A) $x(x + 4) + 6(x + 2)$
(B) $x(x + 6) + 4(x + 2)$
(C) $x(x + 4) + 6(x + 4)$
(D) $x(x + 10) + 2(x + 6)$
(E) $x(x + 6) + 4(x + 6)$

Open Ended Questions (FRQ)

- Q9.** A logistics company has a cost function $C(x) = 3x^2 + 12x + 8x + 32$ where x is the number of employees. Factor $C(x)$ by grouping and verify your answer by multiplication. Provide your work and explanations.
- Q10.** Factor the expression $2x^2 + 10x + 8x + 40$ by grouping. Then, use the factored form to find the values of x for which the expression equals zero. Provide your work and explanations.

Chef's Tips

- Tip 1: Make sure to check students' work for the correct application of factoring by grouping.
- Tip 2: Verify that students are using the correct order of operations when multiplying to check their answers.
- Tip 3: Encourage students to use real-world scenarios, such as transportation and logistics, to demonstrate the practical application of factoring by grouping.